

GIT BRANCHING CONCEPTS & COMMANDS

❖ What is a branch in Git?

- A branch in Git is a **pointer** to a snapshot of changes. It allows parallel **development** by **enabling work** on features, bug fixes, or experiments **without affecting** the main codebase.

❖ Why Use Branches?

- Work on multiple features in parallel
- Experiment safely
- Collaborate without affecting the main codebase

❖ What are the common types of Git branches?

- **main or master:** Main stable production
- **branch feature/:** Used to develop new features
- **bugfix/:** For fixing bugs release/: For preparing releases
- **hotfix/:** For urgent fixes directly in production

❖ Git Branch Commands

- **Create a new branch:**
 - ✓ `git branch feature/login`
- **List branches:**
 - ✓ `git branch`
- **Switch to another branch:**
 - ✓ `git checkout feature/login`
- **Create and switch in one command:**
 - ✓ `git checkout -b feature/login`

❖ git diff – Check Differences

- Shows changes not staged:
 - ✓ `git diff` Shows staged vs last commit:
 - ✓ `git diff --staged`
- Compare two branches:
 - ✓ `git diff branch1..branch2`

❖ Merging Branches

- Merging combines changes from one branch into another.
 - ✓ Example: Merge feature branch into main
 - `git checkout main`
 - `git merge feature/login`

❖ If conflicts occur:

- Git will mark the files
- You must manually resolve them
- ✓ Then use: `git add .`
- ✓ `git commit -m "Resolved conflicts"`

❖ Introduction to Remote Repository (GitHub)

- **What is a Remote Repository?**
- ✓ It's a version of your project hosted on the internet or a network—typically using platforms like GitHub, GitLab, or Bitbucket.
- ✓ **Definition (Interview-Ready):**
- ✓ A **Remote Repository** is a **shared Git repository stored on a remote server** that allows multiple developers to **collaborate** and **synchronize** their local work by pushing and pulling changes.

▪ Why Use Remote Repos?

- Team collaboration
- Backup of local repository
- CI/CD integrations
- Code review and pull requests

❖ Connecting to GitHub

1. Create an account on GitHub Build and ship software on a **single, collaborative platform**
 2. **Create** a new repository (keep it empty for now)
 3. Use HTTPS or SSH to **connect the remote**
- git remote
 - **Link your local repo to GitHub:**
 - ✓ `git remote add origin`
<https://github.com/username/repo-name.git>
 - **Check remotes:**
 - ✓ `git remote -v`

▪ Clone, Pull, Push

- **Clone To copy a GitHub repository to your local machine:**
 - ✓ `git clone` <https://github.com/username/repo-name.git>

▪ Pull

- **To update your local repo with changes from GitHub:**
 - ✓ `git pull origin main`

▪ Push

- **To upload your local commits to GitHub:**
 - ✓ `git push origin main`
 - ✓

❖ Git Configuration

- **Before starting, configure Git:**
- **Initialize local repo**
 - ✓ `git init`
- **# Configure user**

- ✓ `git config --global user.name "John"`
- ✓ `git config --global user.email john@example.com`

- **# Add and commit files**
 - ✓ `git add .`
 - ✓ `git commit -m "Initial commit"`
- **# Add remote**
 - ✓ `git remote add origin https://github.com/john/myrepo.git`
- **# Push to GitHub**
 - ✓ `git push -u origin main`

[GENERAL INTERVIEW QUESTIONS]

Q1. What is a branch in Git?

- A branch in Git is a **pointer to a snapshot** of changes.
- It allows parallel development by enabling work on **features, bug fixes**, or experiments **without** affecting the main codebase

Q2. What are the common types of Git branches?

- **main or master:** Main stable production branch
- **feature/:** Used to develop new features
- **bugfix/:** For fixing bugs
- **release/:** For preparing releases
- **hotfix/:** For urgent fixes directly in production

Q3. How do you create and switch to a new branch?

- `git checkout -b new-branch`

Q4. What does git checkout do?

- `git checkout` is used to **switch branches** or **restore** working directory files to a previous state or commit

Q5. What does git diff show?

- ✓ `git diff` displays the differences between:
 - Working directory and staging area
 - Staged files and the last commit
- ✓ Two branches or commits
- **Examples:**
 - `git diff` :- # unstaged vs working directory
 - `git diff --staged` :_ # staged vs last commit

Q6. What is the difference between git pull and git push?

- **git pull:** Fetches and merges changes from the remote repository into your local branch
- **git push:** Sends your local commits to the remote repository

Q7. What is git init used for?

- `git init` initializes a new local Git repository in the current directory.

Q8. How do you check the current branch and list all branches?

- `git branch` # Lists all local branches

- `git branch -r` # Lists all remote branches

Q9. What is git merge?

- `git merge` **integrates** changes from one branch into another.
- Example:**
- `git checkout main`
- `git merge feature-branch`

Q10. What is a merge conflict and how is it resolved?

- A merge conflict occurs when Git cannot automatically combine code due to **incompatible** changes.
- It must be resolved manually by editing the conflicted files, then using:
 - `git add`
 - `git commit`

Q11. What is a remote repository in Git?

- A remote repository is a **version of your project** hosted on the internet (e.g., GitHub). It allows team **collaboration** and **centralized** access.

Q12. What does git clone do?

- `git clone` creates a local copy of a remote repository.

Q13. How do you check the current branch and list all branches?

- `git branch` # Lists all local branches
- `git branch -r` # Lists all remote branches